H2 ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE NAME	ANSWERS

PDG					PDG INDEX NUMBER			

BIOLOGY
9648/02
Paper 2 Core Paper

17 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any one question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	**80**
1	
2	
3	
4	
5	
6	
7	
Section B	**20**
8 / 9	
Total	**100**

This document consists of **19** printed pages.

Section A

Answer **all** the questions in this section.

1(a) Explain why replication only occurs in a 5' to 3' direction. [2]

Shape of active site of DNA polymerase;
Needs 3'OH end to initiate;
Primers only add at 3';

In a DNA structure, what do 5' and 3' represent. [2]

- 5' represents end (of strand of nucleotide) with 'free' **carbon 5** (on deoxyribose) having phosphate group
- 3' represents end (of strand of nucleotide) with 'free' **carbon 3** (on deoxyribose) having hydroxyl group
- Antiparallel

(b) The nitrogenous bases of the double helix are paired in specific combinations: [3]

adenine (A) with thymine (T)
guanine (G) with cytosine (C)

It was mainly by trial and error that Watson and Crick arrived at this key feature of DNA. At first, Watson imagined that the bases paired like with like—for example, A with A. But this model was inconsistent with the uniform width of the double helix.

Explain why the earlier DNA model would have inconsistent width.

- purines (A and G) are twice as wide as pyrimidines (C and T)
- purines (A and G) are nitrogenous bases with two organic rings
- pyrimidines (C and T) are single ring nitrogenous bases.
- A purine-purine pair is too wide and a pyrimidine-pyrimidine pair too narrow to account for a uniform (2-nm) diameter of the double helix.
- Always pairing a purine with a pyrimidine results in a uniform width.

[Any 3]

(c) DNA replicates semi-conservatively by the two strands separating and forming templates for the synthesis of two new strands.

[2]

In addition to the template provided by the DNA, describe the role of two other types of factors that are required for DNA replication.

- Presence of enzymes to catalyse the processes: (Any one enzyme)
  1. Helicase
     Unzipping of DNA double helix so that DNA molecule can be separated in order to create replicating bubble/fork.
  2. Topoisomerase
     breaking, swiveling and rejoining DNA strands to relieve DNA strains during unwinding
  3. DNA Polymerase
     Addition of free deoxyribonucleotides for elongation of the new DNA strand.
  4. DNA ligase
     form phosphodiester bonds between adjacent deoxyribonucleotides to join the Okazaki fragments.
  5. Primase (*Reject RNA primase N2010 comments*)
     to synthesise the RNA primers to provide free 3'OH for DNA Polymerase to elongate the new DNA strand

- Free deoxyribonucleotides / deoxyribonucleoside triphosphate
    6. to provide building blocks for the synthesis of new DNA strand
    7. to provide free 5' Phosphate group for formation of phosphodiester bond with existing DNA strand.

- RNA primers

provide free 3'OH for DNA Polymerase to elongate the new DNA strand

- Single stranded binding proteins

to prevent the DNA from reannealing so that DNA Polymerase can bind / allow complementary base-pairing

- ATP

When a nucleoside triphosphate links to the sugar phosphate molecule backbone of a growing DNA strand, it losses two of its phosphates as a pyrophosphate molecules. Hydrolysis of the bond between the phosphate groups provides energy for the reaction.

NOTE: Enzymes are consider as ONE group of factors – thus only a maximum of 1 mark will be awarded if students mention more than 1 type of enzymes with explanation.

(d) Just as a DNA strand provides a template for making a new complementary strand during DNA replication, it also can serve as a template for transcription.

Fig. 1.1 is a diagrammatic representation of the process of transcription.

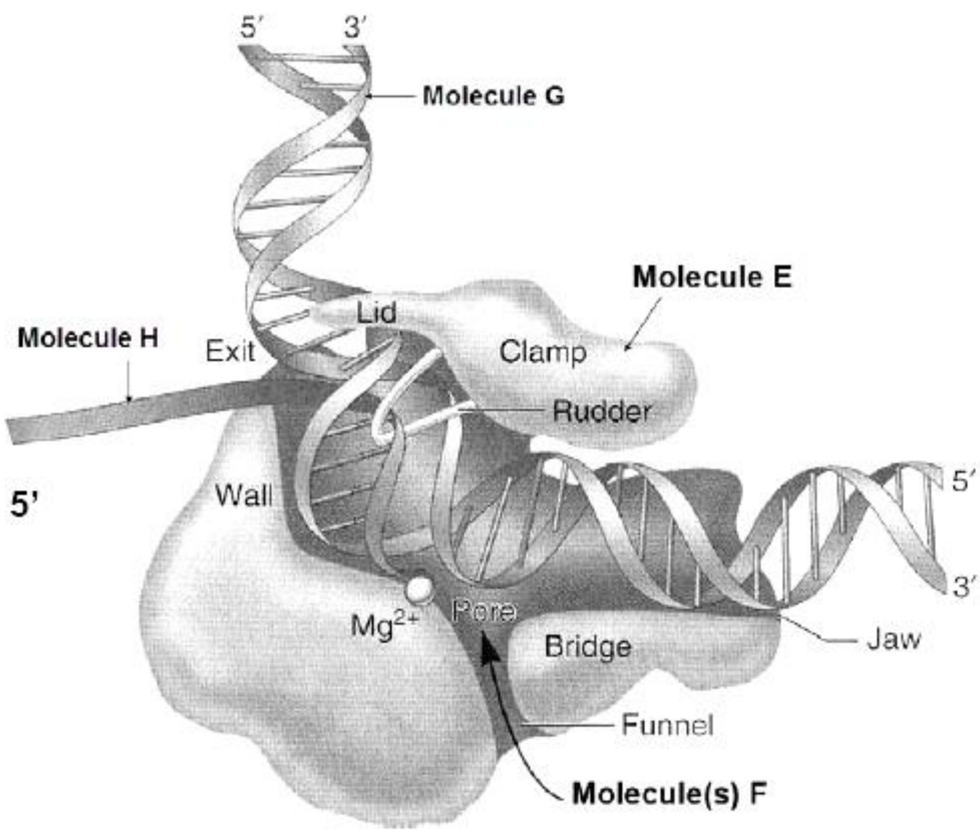

Fig. 1.1

Identify molecule E, F and H. [1]

E – RNA polymerase
F – RNA nucleotides/ Ribonucleoside triphosphate/nucleoside triphosphate
H – mRNA

1m for all 3 correct answers;

(i) Describe how the structure of molecule **E** is adapted to its role in transcription. [3]

1. molecule E has a specific active site which is complementary to substrate such as DNA template, ribonucleotides;
2. R groups of (contact) amino acid residues forms weak (ionic) bonds with substrate;
3. R groups of (catalytic) amino acid residues; that catalyses phosphoester bond formation between growing RNA strand and incoming free nucleotide;
4. funnel allows entry of nucleoside triphosphates;
5. rudder separates the RNA-DNA hybrid;

(ii) Describe one way by which the structure of molecule(s) **F**, which is / are adapted for transcription, differs from that of amino acids, which are adapted for translation. [1]

1. a molecule of F (NTP) is made of a ribose sugar, phosphate group and nitrogenous base (also accept ribulose triphosphate nucleoside (ribose sugar, 3 phosphates, nitrogeneous base); but an amino acid is made of a central α carbon attached to H group, R group, amine group, carboxyl group;

2. identity of molecule F is determined by identity of nitrogenous base; but identity of amino acid is determined by identity of R-group;

3. Molecule F has 4 different types (A, U, G, C) while amino acid has 20 different types (distinct R group).

[Any 1 pt]

[Total : 12]

2 Influenza A viruses are divided into subtypes based on two proteins on the surface of the virus: haemagglutinin (HA) and neuraminidase (NA). The influenza A (H7N9) virus designation of H7N9 identifies it as having HA of the H7 subtype and NA of the N9 subtype.

(a) Describe the roles of haemagglutinin (HA) and neuraminidase (NA) in the reproductive cycle of the influenza (H7N9) virus. [2]

- Haemagglutinin facilitates attachment of virus to host cell membrane by binding to specific sialic acid receptors on epithelial cells.

- Neuraminidase facilitates the release of newly formed virus from the infected host cell by cleaving the sialic acid receptors.

(b) **(i)** The influenza genome comprises genes for RNA polymerase. How is the virus RNA polymerase different from the host RNA polymerase? [1]

- host RNA polymerase is a DNA-dependent RNA polymerase, while the virus RNA polymerase is a RNA-dependent RNA polymerase

(ii) Explain the role of the viral RNA polymerase in the reproductive cycle. [3]

- The viral **RNA polymerase** synthesizes (accept transcribes) complementary ssRNA (+) using ssRNA (-) as a template.

These complementary ssRNA(+) strands serve as:

- mRNAs for eventual translation into viral proteins in the cytoplasm using the host cell ribosomes; and
- templates for making new copies of the ssRNA (-) genome in the nucleus.

(c) H7N9 is an avian influenza virus but it is also able to infect mammals.

Suggest changes that may have occurred to give H7N9 *greater* ability to infect mammals. [2]

- Caused by mutations to haemagglutinin gene during the replication of viral RNA
- Antigenic drift occurs when accumulation of mutations result in a gradual change to haemagglutinin (HA)
- change shape / conformation/ configuration of the haemagglutinin glycoprotein in H7N9
- Shape of haemagglutinin have better fit / more complementary to shape of specific receptors on human host cells

Haemagglutinin of H7N9 was previously only able to bind to sialic acid on epithelial cells of birds. However, H7N9 has now acquired the ability to bind to sialic acid on epithelial cells of mammals. Given that both avian and human influenza strains can infect pigs at the same time, suggest how H7N9 acquired the ability to infect mammals.

Single cell infected with different types of viruses at the same time;
During viral assembly there could be random assembly of RNA segments from different influenza viruses;
Leading to novel gene combinations/genetic shift;

(d) While the influenza genome comprises 8 distinct linear segments of negative sense ssRNA, bacterial genome consists of chromosome and plasmid. Operons are needed for the regulation of gene expression.

Suggest why operons give bacteria a selective advantage. [3]

- Genes of the same metabolic pathways grouped together for control of gene repression/gene switching
- So that bacterium only produces enzymes when required.
- Preventing waste of energy/ resources.
- bacteria able to use a variety of sugars/ substrates as they contain different operons.
- bacteria can respond to environmental change quickly

R description of selective advantage eg. increase in frequency, pass on alleles to next generation.

[Total : 11]

3 In normal person, the fragile X mental retardation protein (FMRP), regulates the synthesis of neurone proteins by stopping ribosomal translocation on target mRNAs. Fig 3.1 shows how patients with fragile X syndrome (FXS) have non-functional FMRP, resulting in accelerated synaptic protein synthesis that leads to abnormal synaptic function and intellectual disability.

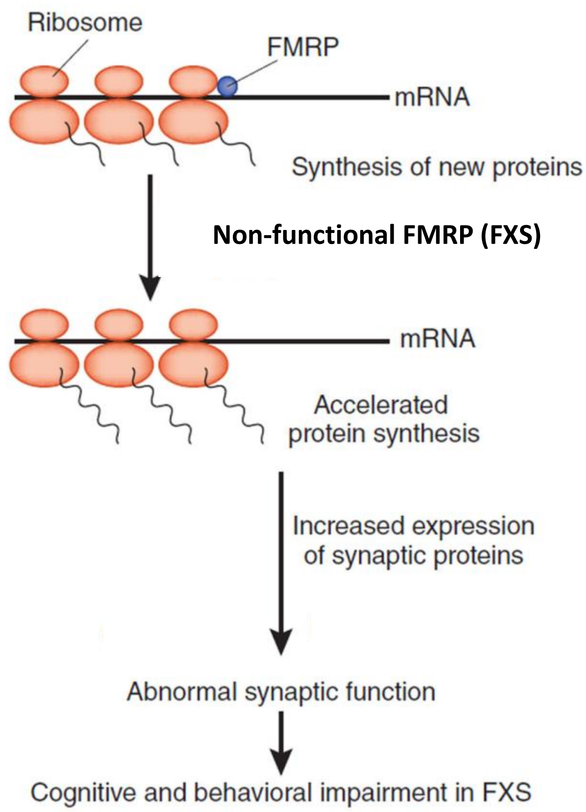

Fig. 3.1

(a) Describe how polysomes are formed.

- Ribosome attaches to **AUG start** codon;
- Another ribosome then attaches to AUG as the preceding ribosome translocates from **5' to 3' direction** of mRNA;

Comments:

- The question asked "how are polysomes formed" and not "what are polysomes" or "why do polysomes form". Hence, answers related to "beads on the string structure", "polysomes are multiple ribosomes on an mRNA", and "they undergo translation simultaneously to speed up translation" are not accepted. [2]

(b) **(i)** State the level of control by FMRP on synaptic protein expression.

- Translational control;

Comments:
This was generally well answered by the majority. Those who got it wrong either identified the wrong stage of control or answered "positive" and "negative" control. [1]

(ii) Describe one other control mechanism of a similar level as **(b)(i)**.

- Presence of translation initiation factors;
- Facilitates the initiation of translation;

OR

- Binding of translational repressors/ regulatory proteins to 5'UTR/ covers AUG;
- Prevents the initiation of translation/ assembly of ribosome;

OR

- Polyadenylation/ Addition of poly(A) tail;
- A longer polyA tail increases the half life of mRNA so more translation;

[2]

(iii) Explain how FMRP stops the ribosome from translocation.

- FMRP has a **specific binding site complementary** in shape to a specific mRNA **base sequence/ codon**;
- Physically **blocks** / steric hindrance to ribosome;

Accept: FMRP has amino acid R groups that attach to mRNA by weak hydrogen/ ionic bonds;

(c) Initial gene therapy trial uses a short stretch of anti-sense RNA. Explain how this will help to reduce the symptom of FXS. [2]

- Anti-sense RNA binds by complementary base pairing;
- Ref to binding to same stretch of mRNA bases that FMRP binds (to cause the same effect of stopping ribosomal translocation);

(d) Synaptic proteins that are folded wrongly are usually degraded.

Suggest the advantages of degrading proteins that are wrongly folded.

- Can **recycle amino acids** for other protein synthesis;
- Regulate cellular processes by **removing enzymes and regulatory proteins** that are no longer needed;
- **Prevent accumulation** of unwanted protein in the cytoplasm;

(Any 2)

Comments:
Some candidates suggested preventing the wrongly folded protein from causing a phenotypic effect. This is not accepted because usually wrongly folded protein is non-functional and would not cause any phenotypic effect. [2]

[Total : 11]

4 Severe combined immunodeficiency (SCID) results from defects in signaling downstream of cytokine receptors that contain the common cytokine-receptor γ-chain (γc). γc is a component of the receptors (R) for many interleukin such as interleukin-2 (IL-2) and IL-7.

Fig. 4.1 shows the events in the JAK-STAT cell signaling pathway when cytokine binds and this in turn leads to an activated receptor. Fig. 4.2 shows two causes that could lead to a non-functional receptor.

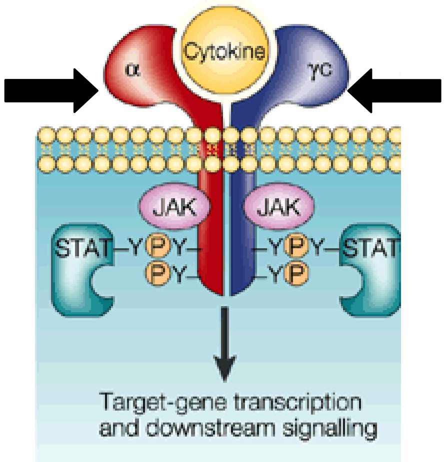

Fig. 4.1

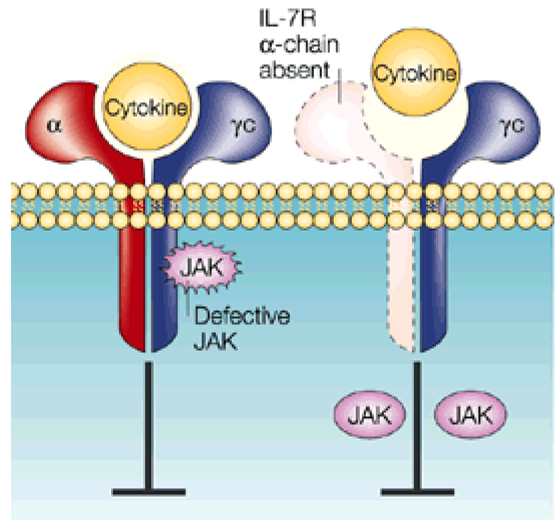

Fig. 4.2

(a) Explain why transmembrane proteins such as the cytokine receptor are especially suited for a role in cell signalling. [2]

- **Spans the entire membrane**;
- with one part interacting with **extracellular** ligand/ signal molecule;
- and another interacting with **intracellular** proteins;
- ligand binds via complementary shape;
- Leading to conformational change of the transmembrane protein/ dimerisation of receptor;
- Cause activation of intracellular enzymes / intracellular proteins inside the cell;
- Allows cellular responses without the external signal having to move into the cell/ able to transduce external signal into internal signal;

(1/2 mark each)

(b) With reference to Fig. 4.1, suggest when happens after cytokine binds to the receptor. [2]

- Dimerization of α and γ chains;
- Activates JAKs;
- (Activated JAK) phosphorylates the intracellular tails of the receptors;
- Which activate/ phosphorylate STATs

(1/2 mark each)

(c) With reference to Fig. 4.2, explain how a change in the sequence of the DNA nucleotide may affect the signaling downstream of cytokine receptors. [5]

Compulsory points:

- Leads to change in mRNA codon leading to a change in amino acid sequence;

2 examples (2m):

- change in amino acid R-group properties, lead to polypeptide chain folded wrongly, resulting in

	a defective JAK • change in amino acid R-group properties, lead to polypeptide chain cannot embed in the membrane, resulting in a absent γc or absent IL-7R α chain • ref to stop codon in mRNA resulting in shorter polypeptide or no polypeptide chain made, resulting in absent γc or absent IL-7R α chain and; relevant consequences (2m): • absent chain in receptor, cytokine cannot bind; • defective JAK, cannot be activated, subsequent events cannot occur;	
(d)	Proteins synthesized by ribosomes attached to the endoplasmic reticulum may be embedded in the cell surface membrane. Outline the route taken by such a protein. • Protein is <u>inserted</u> into the membrane of the rough endoplasmic reticulum; • Vesicles with protein embedded in membrane <u>bud off</u> from membrane of <u>RER;</u> • move to and <u>fuse with cis face;</u> • of <u>Golgi apparatus</u>; • modified and sorted; • Vesicles bud off from the trans face of Golgi apparatus; • <u>fuse</u> with the <u>plasma/cell membrane</u> and receptor is presented to the exterior of the cell; • movement across cell is <u>aided by microtubules of cytoskeleton;</u> (1/2 mark each)	[3]
		[Total: 12]

5 Chickens have a structure called a comb on their heads. The drawings show two types of comb.

Pea comb   Single comb

Fig. 5.1

The shape of the comb is controlled by two alleles of one gene. The allele for pea comb, **A**, is dominant to the allele for single comb, **a**. The colour of chicken eggs is controlled by two alleles of a different gene. The allele for blue eggs, **B**, is dominant to the allele for white eggs, **b**.

The genes for comb shape and egg colour are situated on the same chromosome.

A farmer crossed a chicken that had pea comb and produced blue eggs with a chicken that had a single comb and produced white eggs. The chicken that had a pea comb and produced blue eggs are the offspring of the cross between pure bred pea comb, blue eggs and pure bred single comb, white eggs.

(a) Construct a genetic diagram to illustrate the above cross. [3]

Gametes 1M;
Punnett Square 1M;
F2 phenotype and corresponding genotype ratios 1M;

Parental phenotype	Pea comb, blue eggs	X	Single comb, white eggs
Parental genotype	A B / a b		a b / a b
Gametes	A B, A b, a B, a b		a b

	A B	A b	a B	a b
a b	A B / a b	A b / a b	a B / a b	a b / a b
F_2 genotype ratio	A B	A b	a B	a b

	a b	a b	a b	a b
F_2 phenotype ratio	1 Pea comb blue eggs	1 Pea comb, white eggs	1 single comb, Blue eggs	1 single comb, white eggs

(b) The farmer could identify which of the female offspring from this cross would eventually produce blue eggs. Explain how. [2]

Pea comb offspring will produce blue eggs;
Alleles **A** and **B** are inherited together / are on the same chromosome;

(c) However, in certain crosses, the farmer was not able to identify the female offspring which will produce blue eggs. Explain why. [2]

Crossing over will separate alleles;
If crossing over occurred some gametes will contain alleles **A** and **b;**

In chickens the males are the homogametic sex while the females are the heterogametic sex.

(e) A gene on the X chromosome controls the rate of feather production. The allele for slow feather production, **F**, is dominant to the allele for rapid feather production, **f**.

A farmer made a cross between two chickens with known genotypes. He chose these chickens so that he could tell the sex of the offspring soon after they hatched by looking at their feathers.

Which of the crosses shown in the table did he make? Explain your answer.

Cross	Genotype of male parent	Genotype of female parent
A	$X^F X^F$	$X^f Y$
B	$X^F X^f$	$X^f Y$
C	$X^f X^f$	$X^F Y$
D	$X^F X^f$	$X^F Y$

Fig. 5.3 [2]

- Cross C / $X^f X^f$ and $X^F Y$;
- (Only) cross where all males are one phenotype and all females are a different phenotype;
- Cross showing all males offsprings are slow feather production, all females offsprings fast feather production;

(f) Female chickens are more likely than male chickens to show recessive sex-linked characteristics. Explain why. [2]

- Two alleles for each gene present in male / chromosomes are homologous in male;

OR female has one allele for each gene (for the non-homologous section of chromosome);

- Recessive alleles always expressed in female;
- Males need two recessive alleles for allele to be expressed / in males recessive alleles can be masked by dominant allele

[Total: 11]

6 Table 6.1 shows the approximate percentage of protein in different membranes.

Table 6.1

Organelle	Membrane	Protein as a percentage of dry weight
Chloroplast	Outer membrane	54
Chloroplast	Inner membrane	70
Mitochondrion	Outer membrane	55
Mitochondrion	Inner membrane	80

(a) Describe one way in which the distribution of protein in the inner and outer membranes of these two organelles is similar.

Suggest a reason for this similarity. [2]

Similarity:

- Both the chloroplast and mitochondrion have **more protein** on their inner membrane as compared to outer membrane;

Reason:

Inner membrane of mitochondrion and chloroplast has **various enzymes and proteins such as ATP synthase, proton pumps, electron transport proteins/ electron carriers in ETC** required to carry out chemiosmosis in respiration or photosynthesis;

Note: Chloroplast has a double outer membrane and the inner membrane refers to the thylakoid membrane.

(b) **(i)** Atrazine is a widely used herbicide (weedkiller). It binds to a chloroplast protein involved in electron transfer between photosystems.

Explain how atrazine works as an effective herbicide. [4]

- Atrazine binds to an electron carrier or name a possible electron carrier;
- shape of electron carrier change;
- ETC stop/ photophosphorylation stop;
- no ATP or reduced NADP will be produced and therefore the Calvin Cycle will cease to function;
- no G3P so no source of energy/sugar/food produced

(ii) Maize plants are insensitive to atrazine.

The cytoplasm of maize plant cells contains a tripeptide which binds to atrazine. The product is then transported to the vacuole.

Suggest why maize plants are insensitive to atrazine. [2]

- Atrazine will not enter the chloroplast.
- when in the vacuole the tonoplast will act as an effective barrier preventing the atrazine affecting the metabolism of the cell
- Binding to tripeptide changes configuration of atrazine such that it is not able to enter the chloroplast.

(c) At the end of photosynthesis, glyceraldehyde 3-phosphate (3-carbon sugar) is produced by the Calvin cycle, which is subsequently converted into starch for storage. [3]

Discuss the suitability of starch as a storage compound in plants.

- Large molecule – prevented from diffusing out
- Large molecule made up of a long chain of α-glucoses linked by many glycosidic bonds – able to store large amount of energy
- In a compact space – because it can be coiled into a helix.
- Helical formation was made possible with the OH groups pointing into the helix - insoluble
- Insoluble – thus does not affect the osmotic potential of the cell
- No cross linked/glycosidic bonds on the exterior of the helices – easily hydrolyzed

[Total : 11]

6 Fig 6.1 shows T.W. Engelmann's experiment to measure the action spectrum of the filamentous green alga. He placed the filamentous green alga into a test-tube along with a suspension of oxygen-seeking bacteria. He allowed the bacteria to use up the available oxygen and then illuminated the alga with light that had been passed through a prism to form a spectrum. After a short time he observed the results shown in Fig. 6.1.
(Fig on pg 333)

(a) **(i)** Sketch the action spectrum for this alga. [1]

2 peaks at correct range: one between 400 to 500 range the other between 600 to 700

(ii) Explain his observation shown in Fig. 6.1. [2]

More light energy absorbed at the two range;
By the *light harvesting complexes*;
Increase rate of *photolysis of water*;
To *fill the electron hole* of *chlorophyll a* at reaction centre *PS II* or *PS700*;
Produce *more oxygen* at the two region (to attract oxygen-seeking bacteria)

(b) **(i)** At the end of photosynthesis, glyceraldehyde 3-phosphate (3-carbon sugar) is produced by the Calvin cycle. Outline how this glyceraldehyde 3-phosphate is made in the Calvin cycle and converted into starch. [5]

- (Carbon fixation phase) Carbon dioxide (1C) is accepted by RuBP (5C compound), catalyzed by ribulose bisphosphate carboxylase (Rubisco);
- 6C compound then cleaves to form 2 molecules of 3C glycerate-3-phosphate (GP);
- (reduction phase) in the presence of ATP and NADPH, GP forms 3C triose phsophate (G3P);
- For every 6 G3P formed, 1 G3P exits the cycle
- G3P can form glucose via condensation
- Many α-glucose monomers are linked to form long chain via glycosidic bonds
- Long chain then coils to form compact helix

(ii) Discuss the suitability of starch as a storage compound in plants. [3]

- Large molecule – prevented from diffusing out
- Large molecule made up of a long chain of α-glucoses linked by many glycosidic bonds – able to store large amount of energy
- In a compact space – because it can be coiled into a helix.
- Helical formation was made possible with the OH groups pointing into the helix - insoluble
- Insoluble – thus does not affect the osmotic potential of the cell
- No cross linked/glycosidic bonds on the exterior of the helices – easily hydrolyzed

[Total : 11]

7 The marine threespine sticklebacks, *Gasterosteus aculeatus* is a freshwater fish living in the lakes of British Columbia, Canada as shown in **Fig. 7.1.**

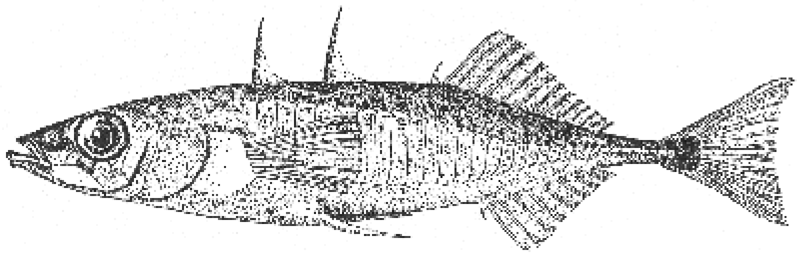

Fig. 7.1

In order to investigate the process of speciation in these populations, three small lakes were studied. Each lake contained two varieties of stickleback: a large, bottom-dwelling variety that fed on invertebrates near the shore and a small, plankton-eating variety that lived in the open water. The probability of breeding between pairs of individuals was measured under laboratory conditions in the following breeding combinations:

I different varieties from the same lake
II different varieties from different lakes
III same variety from different lakes
IV same variety from the same lake

The data are summarized in Fig. 7.1 below.

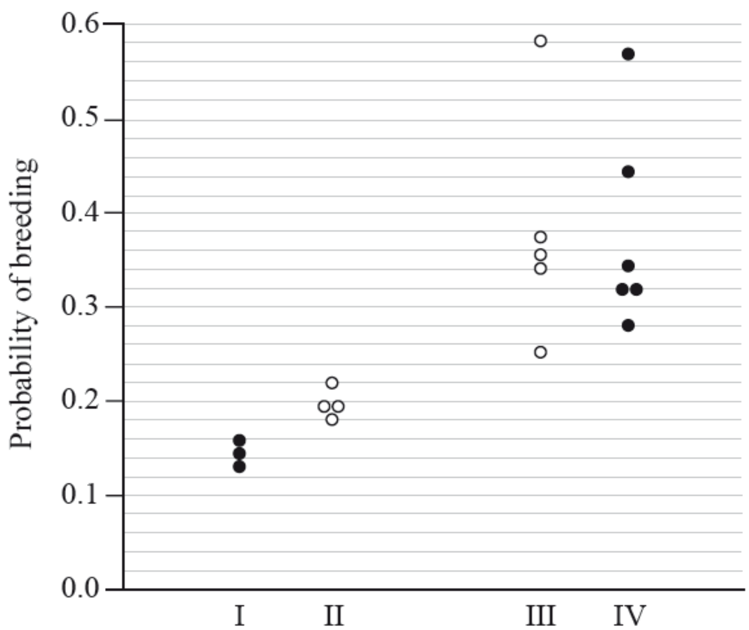

Fig. 7.1

(a) **(i)** Identify the highest and lowest probabilities of breeding for individuals of the same variety from different lakes.

- Highest probability: 0.58 (Allow answers from 0.57–0.59) [1]

Lowest probability: 0.25 (Allow answers from 0.24–0.26)
Both required for the mark.

(ii) With reference to Fig. 7.1, describe the differences in probability of breeding between individuals from the same lake. [2]

- IV combination (of the same variety): Individuals have higher breeding probability if they are the same variety and
- quote values 0.58;

OR

- *Accept:* Individuals of different varieties have a lower probability of breeding and
- quote values;
- The probability of breeding between individuals of the same variety shows a larger range of values / narrower range if of different variety;

[Any 2]

(b) Scientists concluded that speciation is taking place in these populations. Explain the type of speciation as shown in Fig. 7.1.

- **Sympatric speciation**
- is taking place because different varieties from the same lake have a low probability of breeding

OR

- **No evidence of allopatric speciation**
- as same varieties from different lakes do not show strong reproductive isolation/ Comparing exp III and IV, they show similar range of breeding probability.

- **Different feeding habits or habitat** (shore versus open water) contribute to low breeding probability/ reproductive isolation;
- **Reducing interbreeding and gene flow** between the varieties in each lake;

[4]

(c) With reference to Fig. 7.1, explain why all the individuals are still considered the same species.

- **Probability of breeding is not zero/** Not completely reproductively isolated;
- (According to Biological species concept), they can still **interbreed** to produce **fertile, viable offspring**; [2]

(d) The freshwater lakes also contain many different types of parasites that infect the marine threespine sticklebacks. Suggest why these parasites help to speed up speciation of the marine threespine sticklebacks.

- Parasites are **host specific**/ infect either bottom-dwelling variety or plankton-eating variety;
- Parasites constitute a **selection pressure**;
- Cite an example of a characteristic that confer a selective advantage, e.g. resistance against parasitic infection
- Those selected for survive, reproduce and pass on their alleles coding for the beneficial characteristic to their offsprings. [3]

[Total : 12]

Section B

Answer **one** question.
Write your answers on the separate answer paper provided.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section (a), (b) etc., as indicated in the question.

8 **(a)** Explain how the positive and negative control of the lac operon affect bacteria growth. [10]

[Compulsory 1 point]

- Metabolism is biased toward the utilization of glucose - glucose is the simplest form of sugar /enzymes required for the breakdown of glucose via glycolysis are continuously present.

[Max 3m]

- When [glucose] high and [lactose] is absent, glucose is preferred over lactose/used as a main respiratory substrate/ main energy source.
- Repressor binds to operator;
- [cAMP] decreases, CAP is not activated/cAMP does not bind to CAP and CAP unable to bind to CAP binding site;
- RNA polymerase does not bind to promoter/ lac operon is switched off;
- bacteria growth increases exponentially because energy from glucose breakdown is utilized for bacterial growth.

[Compulsory 2 points]

- When glucose is fully utilized, bacterial growth will remain constant
- [lactose] also remains constant initially as time is needed for lac operon to turn on genes to produce ß-galactosidase to hydrolyse lactose;

[Max 3m]

- Absence of [glucose] and high [lactose];
- Allolactose acts an inducer, binds to repressor to inactivate it;
- high levels of cAMP binds and activates CAP and
- increases affinity of RNA polymerase to bind to promoter and to turn on expression of lac genes;
- high [lactose] - Bacteria growth increases exponentially due to production of ß-galactosidase that hydrolyses lactose to form glucose (main respiratory substrate) and galactose;
- as [lactose] concentration decreases; less allolactose binds to the repressor; and as a result, the activated repressor can bind to the operator;
- this blocks binding of RNA Polymerase to the promoter and inhibits transcription of lac operon;
- bacteria growth plateaus/drops as lactose is used up

Description for bacteria growth [Compulsory 3m]

- bacteria growth increases exponentially because energy from glucose breakdown is utilized for bacterial growth.
- When glucose is fully utilized, bacterial growth will remain constant
- high [lactose] - Bacteria growth increases exponentially due to production of ß-galactosidase that hydrolyses lactose to form glucose (main respiratory substrate) and galactose;
- bacteria growth plateaus/drops as lactose is used up

(b) Describe how regulation of blood glucose level in humans involves cAMP. [5]

- α-cells detects low blood glucose level,
- secrete glucagon into bloodstream;
- glucagon binds to GPCR at the cell surface membrane, activates GPCR;
- Activated GPCR binds to G-protein;
- Cause GDP to be displaced by GTP, activates G-protein;
- Activated G-protein activated adenylyl cyclase;
- Converts ATP to cAMP;
- Which acts as 2nd messenger
- Activates phosphorylation cascade
- Activate glycogen phosphorylase
- Break down glycogen to glucose (to increase blood glucose level)

(1/2 m each)

(c) Explain the significance of the steps in glycolysis. [5]

- Glycolysis is a common step in both anaerobic and aerobic respiration;
- Phosphorylation of glucose (by 2 ATP) is to activate it;
- Phosphorylation of glucose (by 2 ATP) / glucose-6-phosphate results in glucose being trapped in the cytosol/ unable to leave the cell through the same glucose carrier protein/ committed to the end of glycolysis;
- PFK which catalyse the phosphorylation of fructose phosphate also control rate of glycolysis/ high rate of ATP act as allosteric inhibitor to PFK;
- ATP synthesis by substrate level phosphorylation;
- Forms two glyceraldehyde-3-phosphate/ triose phosphate from one glucose;
- Forms two reduced NAD (NADH) (by dehydrogenation);
- Which later give 6 ATP by oxidative phosphorylation;
- Pyruvate can enter into link reaction/ mitochondria or be converted to lactate (in mammals) or ethanol and carbon dioxide (in yeast);
- Pyruvate is small enough to enter mitochondrion for aerobic respiration;

9 **(a)** Describe the differences between cellulose synthase and cellulose. [5]

Any five below:

Feature	Cellulose synthase	Cellulose
Nature	Globular protein	Polysaccharide
Monomer	Amino acid	β glucose
Bond between monomer	Peptide bond	β (1,4) glycosidic bond
Shape	Spherical	Linear (chains)
Bonds contributing to structure	Ionic bond, hydrogen bond, hydrophobic interaction and disulfide bonds	Hydrogen bonds
Groups contributing to bond formation	between R groups	between OH groups
Location/ Function	On cell surface membrane/ enzyme	As part of plant cell wall surrounding plant cell/ gives cell a regular shape

(b) With reference to photosynthesis and cellular respiration, explain the importance of compartmentalization within a plant cell. [10]

- The inner membrane of the chloroplast encloses the stroma where enzymes for Calvin cycle are
- Thylakoid membrane houses light harvesting complexes / photosystems for harnessing light energy to excite electrons in the reaction centre down the ETC
- with NADP reductase as final electron carrier which is reduced to NADPH for Calvin cycle
- Flow of electrons down ETC allows pumping of H^+ from the stroma to the thylakoid space, setting up a proton gradient
- Diffusion of the H^+ from the thylakoid space to the stroma via ATP synthase / stalked particles in thylakoid membrane allows phosphorylation of ADP to form ATP required in Calvin cycle
- The inner mitochondrial membrane encloses the matrix where enzymes of the Krebs cycle are localised
- Enzymes localised in the cytoplasm allow glycolysis to occur
- The inner mitochondrial membrane is the site of oxidative phosphorylation
- and is highly folded forming cristae to increase surface area for more ETCs / stalked particles
- Flow of electrons down ETC allows pumping of H^+ from the matrix to the intermembrane space, setting up a proton gradient since the inner mitochondrial membrane is impermeable to H^+
- Diffusion of the H^+ from the inner mitochondrial space to the matrix via ATP synthase / stalked particles in the inner mitochondrial membrane allows phosphorylation of ADP to form ATP

(c) Explain the role of meiosis in natural selection. [5]

(Any 3)

- Crossing over at prophase I (exchanges alleles);
- Independent assortment of homologous chromosomes at metaphase I;
- Introduces genetic variation in a population;
- Random fusion of gametes after meiosis (further increases genetic variation in individuals);

(Compulsory 2 marks)

- Selection pressure results in some organisms having selective advantage;
- Those selected for survive and reproduce and pass on their alleles to their offspring, increasing frequency of alleles over time;